

Andrew Gonzales

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SUMMARY

Embedded Software Engineer with 2+ years in embedded firmware and application software for industrial motion control: EtherCAT slave firmware on SSC-generated stacks (Infineon XMC4300/XMC4800), EtherNet/IP, DMA-driven real-time communication, closed-loop PID tuning for voice-coil and brushless actuators, plus C#/.NET and full-stack JavaScript applications, shipped in production controllers used by industrial customers and European research programs.

EXPERIENCE

Embedded Software Engineer

June 2024 – Present

SMAC Moving Coil Actuators

Carlsbad, CA

- Extended the SSC-generated EtherCAT slave stack on Infineon XMC4300/XMC4800 (Cortex-M4, integrated ESC) with File over EtherCAT (FoE), integrating mailbox handling into the VLC-ETC application layer for acyclic parameter updates and PID tuning data transfer.
- Configured DMA for the Distributed Clock (DC) process data path and integrated it with the stack's SYNC0 handling to meet hard real-time deadlines; adapted the stack to bridge EtherCAT process data to the SMAC motion controller over UART.
- Firmware ships in production VLC-ETC-402 controllers used by industrial customers and European research programs (recent 21-unit, ~\$30K order).
- Modified and validated EtherCAT Slave Information (ESI) files, CoE object dictionary entries, and cyclic/acyclic PDO mappings for TwinCAT 3 integration.
- Measured and validated EtherNet/IP cyclic latency across a five-node PLC chain and contributed to a new IO-based control method that reduced per-unit cost for customers.
- Developed C# .NET desktop applications (WinForms/WPF) for actuator control, automated data capture, and diagnostics: VLC-X Editor, Thread Check GUI, SMAC Simple GUI, VLC-ETC-402 Configuration Tool, and VLC-EIP Configuration Tool.
- Validated actuators across multiple platforms: phasing, HIPOT, closed-loop PID tuning, soft-land, force/velocity/position profiling, cycle testing, and high-resolution data capture.

Associate Software Engineer

August 2025 – February 2026

FRiNGE

Remote (Oceanside, CA)

- Built and extended a full-stack internal admin web application using JavaScript, PostgreSQL, Supabase, and RESTful APIs to manage regions, locations, and media assets.
- Implemented geospatial features with the Google Maps API, rendering location markers and region geofences from stored polygon data.
- Added interactive front-end behavior including hover-based marker scaling and conditional polygon rendering.

PROJECTS

Pan/Tilt Laser Turret Firmware | STM32F411, C, HAL, GNU Make, J-Link | github.com/OhitsAndrewG/pan_tilt

Sept 2026 – Present

- Bare-metal, fully non-blocking interrupt-driven firmware: UART at 115200 with a 256-byte TX ring buffer and line-framed, bounds-checked RX; HC-SR04 ranging via timer input capture at 1 μ s resolution with echo timeout and 10 Hz pacing; laser via NPN low-side driver; KEY:VALUE serial protocol with OK/ERR acknowledgement and telemetry to a Raspberry Pi host.
- Firmware built on STM32 HAL/CMSIS with a CubeMX-generated base; GNU Make build with a GNUMakefile wrapper adding flash, debug, and telemetry targets; SWD debugging with J-Link, GDB, and VS Code Cortex-Debug. Serial link bridges to a Raspberry Pi 4 host that carries the camera; servo PWM control and Pi host software in progress.

Spotify Now-Playing LED Matrix Display | Python, Raspberry Pi Zero 2 W, rgbmatrix, Pillow, systemd

2026

- Built an always-on display rendering live Spotify playback (album art, scrolling title, elapsed timer, progress bar) at 20 fps on a 64x32 RGB LED matrix driven by a Raspberry Pi Zero 2 W, with a custom 3x5 bitmap font and double-buffered vsync rendering; deployed headless as a systemd service.
- Two-thread architecture separating Spotify Web API polling from the 20 fps render loop with lock-guarded shared state; client-side playback position interpolation between 5s polls; OAuth 2.0 refresh-token authentication with proactive token renewal; Retry-After rate-limit backoff and fallback to a clock display when idle or offline.

EDUCATION

University of Southern California

Los Angeles, CA

Bachelor of Science in Computer Science

May 2024

TECHNICAL SKILLS

Languages: C, C++, C#, Python, JavaScript, TypeScript, Java, Swift, SQL, HTML, CSS

Embedded: ARM Cortex-M, STM32 (HAL, CMSIS, bare-metal), embedded C, DMA, interrupt-driven design, timers/PWM/input capture, real-time control, PID / closed-loop control, UART, SPI, I2C, JTAG/SWD, STM32CubeMX, ST-Link, J-Link

Industrial Protocols: EtherCAT (SSC slave stack, CoE, FoE, Distributed Clocks, ESI, Infineon XMC ESC), EtherNet/IP, TwinCAT 3, Logix 5000, HTTP/REST

Software: C# .NET (WinForms, WPF), React, Next.js, Express.js, Spring Boot, SwiftUI, REST APIs, OAuth 2.0, multithreading, PostgreSQL, Supabase, Firebase, Pillow, NumPy, Pandas, Matplotlib, OpenCV

Tools: Git, GNU Make, CMake, GDB, VS Code Cortex-Debug, Docker, Linux (Debian, systemd, SSH), Windows, macOS